MKP PCB 用 DC-Link 电容器 105°C

MKP DC-Link Capacitor for PCB 105°C



系列 Series MKP

行业型号 Industry code C3D

产品类型 Type

PCB 用 DC-Link 电容器
DC-Link Capacitor for PCB

应用 Applications

- 高性能直流滤波应用场合(如：变频器、工业和高端电源、太阳能逆变器等)
  High performance DC filtering applications (i.e. transducers, industrial and high-end power supplies and solar inverters)

技术要求 Specifications

项目 Item	参数 Specification
引用标准 Reference Standard	IEC 61071 (GB/T 17702)
气候类别 Climatic Category	40/85/56
工作温度范围 Operating Temperature Range	-40°C~+105°C (+85°C to +105°C: decreasing factor 1.35% per °C for $U_{N,85°C}$)
额定电压 V_R (70°C) → 85°C 工作电压 V_{op} Rated voltage V_R (70°C) → operating voltage V_{op} (85°C)	450→450Vdc, 575→500Vdc, 800→700Vdc, 900→800Vdc, 1100→920Vdc, 1300→1100Vdc（成对对应，见“额定电压与 85°C 工作电压”表 / paired, see voltage table）
电容量偏差 Capacitance Tolerance	±5%(J), ±10%(K)
耐电压 Voltage proof	1.5U_N (10s)
绝缘电阻 Insulation Resistance (IR×C_N)	⩾10000s (20°C, 100Vdc, 1min)
自感 (Ls) Self Inductance (Ls)	<1nH per mm of lead spacing
最大峰值电流 I(A) Maximum peak current	I=C*dV/dt
预期寿命 Expected lifetime	100000h @U_N and θhs=70°C

额定电压与 85°C 工作电压 Rated voltage and 85°C operating voltage

V_R 为 70°C 下的额定电压；在 85°C 使用时须降额至对应的工作电压 V_{op}（85°C 以上至 105°C 再按每°C 1.35% 对 V_{op} 降额）。
V_R is the rated voltage at 70°C; at 85°C the voltage is derated to the paired operating voltage V_{op} (above 85°C up to 105°C, derate V_{op} by a further 1.35% per °C).

额定电压 V_R (70°C) / Rated voltage V_R at 70°C	85°C 工作电压 V_{op} / Operating voltage V_{op} at 85°C
450 Vdc	450 Vdc
575 Vdc	500 Vdc
800 Vdc	700 Vdc
900 Vdc	800 Vdc

额定电压 V_R (70°C) / Rated voltage V_R at 70°C	85°C 工作电压 V_{op} / Operating voltage V_{op} at 85°C
1100 Vdc	920 Vdc
1300 Vdc	1100 Vdc

订货规格 Ordering Specifications

订货规格 引脚间距 27.5 mm，$V_{R,70°C}$ = 450 V DC, $V_{op,85°C}$ = 450 V DC Ordering codes and packing units (lead spacing 27.5 mm), $V_{R,70°C}$ = 450 V DC, $V_{op,85°C}$ = 450 V DC

C_R [1] (µF)	Max. dimensions W×H×T (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
5.0	31.5 x 21.0 x11.0	5.0	21.1	19.0	1.2	10.7
10.0	31.5 x 24.5 x15.0	8.0	10.9	24.0	1.2	11.0
22.0	31.5 x 36.5 x22.0	14.5	5.4	30.0	1.3	12.1

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。
[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 27.5 mm，$V_{R,70°C}$ = 800 V DC, $V_{op,85°C}$ = 700 V DC Ordering codes and packing units (lead spacing 27.5 mm), $V_{R,70°C}$ = 800 V DC, $V_{op,85°C}$ = 700 V DC

C_R [1] (µF)	Max. dimensions W×H×T (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
3.0	31.5 x 21.0 x11.0	4.5	24.8	19.0	0.9	7.6
5.0	31.5 x 24.5 x14.0	6.5	15.3	23.0	0.9	7.7
12.0	31.5 x 36.5 x22.0	13.0	6.8	34.0	1.0	8.3

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。
[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 27.5 mm，$V_{R,70°C}$ = 1100 V DC, $V_{op,85°C}$ = 920 V DC Ordering codes and packing units (lead spacing 27.5 mm), $V_{R,70°C}$ = 1100 V DC, $V_{op,85°C}$ = 920 V DC

C_R [1] (µF)	Max. dimensions W×H×T (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
2.0	31.5 x 21.5 x12.5	4.5	26.3	19.0	0.7	5.3
3.3	31.5 x 27.5 x18.0	7.0	16.2	22.0	0.7	5.4
5.0	31.5 x 30.0 x19.0	9.0	10.9	27.0	0.7	5.5
7.0	31.5 x 36.5 x22.0	12.0	8.1	30.0	0.7	5.8

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。
[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 27.5 mm，$V_{R,70°C}$ = 1300 V DC, $V_{op,85°C}$ = 1100 V DC Ordering codes and packing units (lead spacing 27.5 mm), $V_{R,70°C}$ = 1300 V DC, $V_{op,85°C}$ = 1100 V DC

C_R [1] (µF)	Max. dimensions W×H×T (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
1.5	31.5 x 21.5 x12.5	4.4	31.3	20.0	0.6	4.8
3.0	31.5 x 27.5 x18.0	7.0	16.0	24.0	0.6	4.9
5.0	31.5 x 36.5 x22.0	10.5	9.8	33.0	0.7	5.1

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。
[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 37.5 mm，$V_{R,70°C}$ = 450 V DC, $V_{op,85°C}$ = 450 V DC Ordering codes and packing units (lead spacing 37.5 mm), $V_{R,70°C}$ = 450 V DC, $V_{op,85°C}$ = 450 V DC

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
30.0	42.0 x 39.5 x 20.0	10.2	14.0	7.0	11.0	2.3	21.3
30.0	42.0 x 39.5 x 20.0	-	14.0	7.3	28.0	2.4	22.3
35.0	42.0 x 37.0 x 28.0	10.2	16.5	6.0	10.0	2.3	21.4
35.0	42.0 x 37.0 x 28.0	-	16.0	6.4	24.0	2.4	22.6
40.0	42.0 x 37.0 x 28.0	10.2	17.5	5.3	11.0	2.3	21.4
40.0	42.0 x 37.0 x 28.0	-	17.0	5.6	26.0	2.4	22.7
50.0	42.0 x 42.5 x 28.0	10.2	20.0	4.3	12.0	2.3	21.7
50.0	42.0 x 42.5 x 28.0	-	19.0	4.7	30.0	2.5	23.8
60.0	42.0 x 45.0 x 30.0	20.3	23.5	3.6	14.0	2.4	22.3
60.0	42.0 x 45.0 x 30.0	-	22.0	4.0	32.0	2.5	24.2
65.0	42.0 x 48.0 x 33.0	20.3	25.5	3.3	14.0	2.3	22.2

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。
[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 37.5 mm，$V_{R,70°C}$ = 575 V DC, $V_{op,85°C}$ = 500 V DC Ordering codes and packing units (lead spacing 37.5 mm), $V_{R,70°C}$ = 575 V DC, $V_{op,85°C}$ = 500 V DC

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
25.0	42.0 x 39.5 x 20.0	10.2	14.0	7.0	12.0	1.9	17.5
25.0	42.0 x 39.5 x 20.0	-	13.5	7.4	28.0	2.0	18.3
30.0	42.0 x 37.0 x 28.0	10.2	16.5	5.8	11.0	1.9	17.6
30.0	42.0 x 37.0 x 28.0	-	16.5	6.1	26.0	2.0	18.5
35.0	42.0 x 42.5 x 28.0	10.2	19.0	5.0	12.0	1.9	17.8
35.0	42.0 x 42.5 x 28.0	-	18.0	5.3	29.0	2.0	19.0
45.0	42.0 x 45.0 x 30.0	20.3	22.0	4.0	13.0	1.9	17.9
45.0	42.0 x 45.0 x 30.0	-	21.0	4.4	32.0	2.1	19.7
50.0	42.0 x 48.0 x 33.0	20.3	25.0	3.5	14.0	2.0	18.1

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。
[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 37.5 mm，$V_{R,70°C}$ = 800 V DC, $V_{op,85°C}$ = 700 V DC Ordering codes and packing units (lead spacing 37.5 mm), $V_{R,70°C}$ = 800 V DC, $V_{op,85°C}$ = 700 V DC

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
14.0	41.5 x 32.5 x 18.0	-	10.0	11.5	23.0	1.8	16.3
15.0	42.0 x 39.5 x 20.0	10.2	12.0	9.6	10.0	1.7	15.2
15.0	42.0 x 39.5 x 20.0	-	11.5	10.3	24.0	1.7	15.7
20.0	42.0 x 37.0 x 28.0	10.2	14.5	7.5	10.0	1.7	15.3
20.0	42.0 x 37.0 x 28.0	-	14.5	7.8	24.0	1.7	15.9
22.0	42.0 x 37.0 x 28.0	10.2	15.5	6.8	11.0	1.7	15.3
22.0	42.0 x 37.0 x 28.0	-	15.0	7.1	25.0	1.7	16.0
25.0	42.0 x 42.5 x 28.0	10.2	17.0	6.1	11.0	1.7	15.4
25.0	42.0 x 42.5 x 28.0	-	16.5	6.4	28.0	1.8	16.3
30.0	42.0 x 45.0 x 30.0	20.3	19.5	5.1	12.0	1.7	15.6
30.0	42.0 x 45.0 x 30.0	-	19.0	5.5	30.0	1.8	16.7

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
35.0	42.0 x 48.0 x 33.0	20.3	22.0	4.3	14.0	1.7	15.7

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。

[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 37.5 mm，$V_{R,70°C}$ = 900 V DC, $V_{op,85°C}$ = 800 V DC Ordering codes and packing units (lead spacing 37.5 mm), $V_{R,70°C}$ = 900 V DC, $V_{op,85°C}$ = 800 V DC

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
15.0	42.0 x 39.5 x 20.0	10.2	12.5	9.1	12.0	1.5	13.6
15.0	42.0 x 39.5 x 20.0	-	12.0	9.4	28.0	1.5	14.1
20.0	42.0 x 37.0 x 28.0	10.2	15.0	7.0	11.0	1.5	13.6
20.0	42.0 x 37.0 x 28.0	-	15.0	7.3	26.0	1.6	14.2
22.0	42.0 x 42.5 x 28.0	10.2	17.0	6.3	12.0	1.5	13.7
22.0	42.0 x 42.5 x 28.0	-	16.5	6.6	29.0	1.6	14.5
25.0	42.0 x 45.0 x 30.0	20.3	19.0	5.5	13.0	1.5	13.8
25.0	42.0 x 45.0 x 30.0	-	18.5	5.9	32.0	1.6	14.7
30.0	42.0 x 48.0 x 33.0	20.3	21.5	4.7	14.0	1.5	13.9

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。

[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 37.5 mm，$V_{R,70°C}$ = 1100 V DC, $V_{op,85°C}$ = 920 V DC Ordering codes and packing units (lead spacing 37.5 mm), $V_{R,70°C}$ = 1100 V DC, $V_{op,85°C}$ = 920 V DC

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
12.0	42.0 x 39.5 x 20.0	10.2	12.0	10.2	12.0	1.4	12.2
12.0	42.0 x 39.5 x 20.0	-	11.5	10.5	28.0	1.4	12.6
14.0	42.0 x 37.0 x 28.0	10.2	13.5	8.7	21.0	1.4	12.2
14.0	42.0 x 37.0 x 28.0	-	13.5	9.0	25.0	1.4	12.6
16.0	42.0 x 42.5 x 28.0	10.2	15.5	7.4	12.0	1.4	12.3
16.0	42.0 x 42.5 x 28.0	-	15.0	7.8	30.0	1.4	12.9
20.0	42.0 x 45.0 x 30.0	20.3	18.0	6.0	14.0	1.4	12.4
20.0	42.0 x 45.0 x 30.0	-	17.5	6.5	32.0	1.4	13.1
22.0	42.0 x 48.0 x 33.0	20.3	21.0	4.9	15.0	1.3	11.4

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。

[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

订货规格 引脚间距 37.5 mm，$V_{R,70°C}$ = 1300 V DC, $V_{op,85°C}$ = 1100 V DC Ordering codes and packing units (lead spacing 37.5 mm), $V_{R,70°C}$ = 1300 V DC, $V_{op,85°C}$ = 1100 V DC

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
8.0	42.0 x 39.5 x 20.0	10.2	11.0	12.1	12.0	1.1	9.7
8.0	42.0 x 39.5 x 20.0	-	10.5	12.4	24.0	1.2	10.0
10.0	42.0 x 37.0 x 28.0	10.2	13.0	9.6	11.0	1.1	9.7
10.0	42.0 x 37.0 x 28.0	-	12.5	9.9	26.0	1.2	10.0
12.0	42.0 x 42.5 x 28.0	10.2	14.5	8.1	12.0	1.1	9.8
12.0	42.0 x 42.5 x 28.0	-	14.0	8.5	28.0	1.2	10.1
14.0	42.0 x 45.0 x 30.0	20.3	17.0	6.8	14.0	1.1	10.1

C_R [1] (μF)	Max. dimensions W×H×T (mm)	P1 (mm)	$I_{RMS,max}$ [2] 70°C 10kHz (A)	ESR_{typ} 70°C 10kHz (mΩ)	ESL_{typ} [3] (nH)	tanδ 1kHz (10^{-3})	tanδ 10kHz (10^{-3})
14.0	42.0 x 45.0 x 30.0	-	16.5	7.3	32.0	1.2	10.4
16.0	42.0 x 48.0 x 33.0	20.3	19.0	6.0	15.0	1.1	9.9

[1] 电容量在 1 kHz 测量；[2] 70°C、10 kHz、ΔT≤20°C、ΔESR_{typ}≤±5% 时的最大纹波电流；[3] 在谐振频率测得的典型 ESL。

[1] Capacitance measured at 1 kHz; [2] max. ripple current at 70°C, 10 kHz, ΔT≤20°C, ΔESR_{typ}≤±5%; [3] typical ESL measured at resonance frequency.

备注 Notes

- 订货编码容差代码：J=±5%、K=±10%；引脚长度 6±1 mm。
  Ordering code tolerance: J=±5%, K=±10%; lead length 6±1 mm.
- 引脚间距 27.5 mm（2-pin）与 37.5 mm（2-pin / 4-pin）两种；27.5 mm 规格未提供 575/500 Vdc 与 900/800 Vdc 两组。
  Lead spacing 27.5 mm (2-pin) or 37.5 mm (2-pin / 4-pin); the 575/500 Vdc and 900/800 Vdc groups are offered in 37.5 mm only.
- **产品标识示例 Marking example:** BKW MKP DC Link / 14μF±10% 800VDC / IEC61071 PJ0601